AI.105

Natural Language Proce

COURSE OVERVIEW

Computational processing of human language: text representation, linguistic structure, and introductory language understanding.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: LANGUAGE FOR MACHINES

KEY TOPICS

- What is NLP
- Tokenization, stemming, lemmatization
- Corpus and text preprocessing
- NLP in AI assistants and robots

TEACHING FOCUS

Turn raw text into structured linguistic data.

KEY TAKEAWAY Language is data with structure and meaning.

2

SESSION

60 MINUTES

TEXT REPRESENTATION

KEY TOPICS

- Bag-of-words and n-grams
- TF-IDF weighting
- One-hot encoding
- Intro to word embeddings

TEACHING FOCUS

Represent words and documents numerically.

KEY TAKEAWAY Meaning can be represented as vectors.

3

SESSION

60 MINUTES

LINGUISTIC STRUCTURE

KEY TOPICS

- Part-of-speech tagging
- Named entity recognition
- Parsing and syntax trees
- Morphology basics

TEACHING FOCUS

Analyze the grammar of sentences.

KEY TAKEAWAY Structure disambiguates meaning.

4

SESSION

60 MINUTES

LANGUAGE MODELING

KEY TOPICS

- N-gram language models
- Probability of word sequences
- Perplexity and smoothing
- Neural language models intro

TEACHING FOCUS

Predict what word comes next.

KEY TAKEAWAY Modeling sequences models language.

5

SESSION

60 MINUTES

NLP APPLICATIONS

KEY TOPICS

- Machine translation
- Sentiment analysis
- Chatbots and dialogue
- Speech recognition intro

TEACHING FOCUS

Deploy language understanding in products.

KEY TAKEAWAY NLP powers human–machine conversation.

NATURAL LANGUAGE PROCESSING

6

SESSION

 60 MINUTES

NLP FOR ROBOTICS & ETHICS

KEY TOPICS

- Human–robot dialogue
- Bias in language models
- Course recap and assessment
- Toward HRI (AI.110)

TEACHING FOCUS

Enable robots to understand and speak.

▲ **KEY TAKEAWAY** Robots must understand humans, not just commands.